

Context Aware To Do app

PRIORIS



21CSC203P

Advance Programming Practice

Introduction

Most existing to-do applications only work with fixed reminders based on date and time. This often fails because people's routines change, and important tasks get missed. Our idea is to build a context-aware to-do app that adapts to the user's situation. Instead of showing every pending task, it uses context like location, time, or activity to display only the most relevant ones. This ensures that users are reminded of the right task at the right moment, making task management smarter and more efficient.

Objectives

- Build a desktop-based context-aware to-do app
- Integrate Eisenhower Matrix for task prioritization
- Provide automatic task suggestions using context triggers
- Simple, minimal interface for better productivity

Methodology

Tag System

- Organizes tasks by context such as location, time, commute, or focus mode
- Uses tags like Home, Work, or Urgent for easy filtering
- Sends smart notifications only when relevant
- Displays Suggested Tasks automatically based on context
- Ensures users see what matters most at the right time

Weather alert!
Get the watch fixed

Looks like it may rain for the next few days, wrap up your pending outdoor tasks before it starts.

Task deadline

Submit assignment- EEE

Your task deadline is approaching.
Finish it soon to stay on track

Suggested Tasks

Weather alert!

Get the watch fixed

Looks like it may rain for the next few days, wrap up your pending outdoor tasks before it starts.

Task deadline

Submit assignment- EEE

Methodology

Eisenhower Matrix

- Prioritizes tasks into four quadrants:
 - Urgent & Important
 - Important but Not Urgent
 - Urgent but Not Important
 - Neither Urgent nor Important
- Helps focus on what truly matters
- Simplifies decision-making
- Guides you on which tasks to act on, schedule, delegate, or drop

1 Urgent and Important

- ☐ Get groceries from market today
- ☐ Submit assignment : EEE tom
- ☐ Pay the electricity bill 12 Sep

2 Not urgent and Important

- ☐ Complete the practical file 16 Sep
- ☐ Organize photo albums on laptop
- ☐ Learn basic cooking recipe

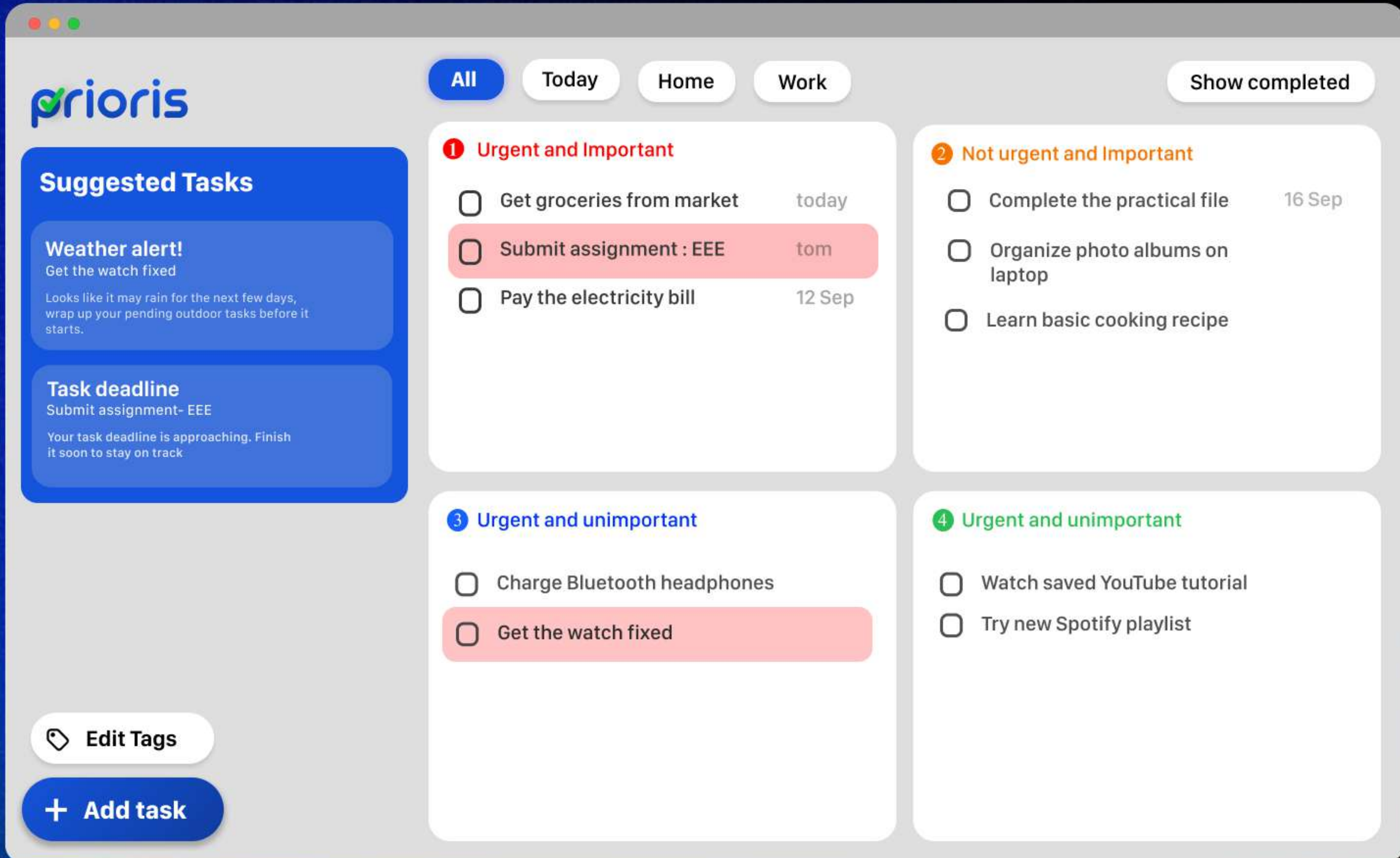
3 Urgent and unimportant

- ☐ Charge Bluetooth headphones
- ☐ Get the watch fixed

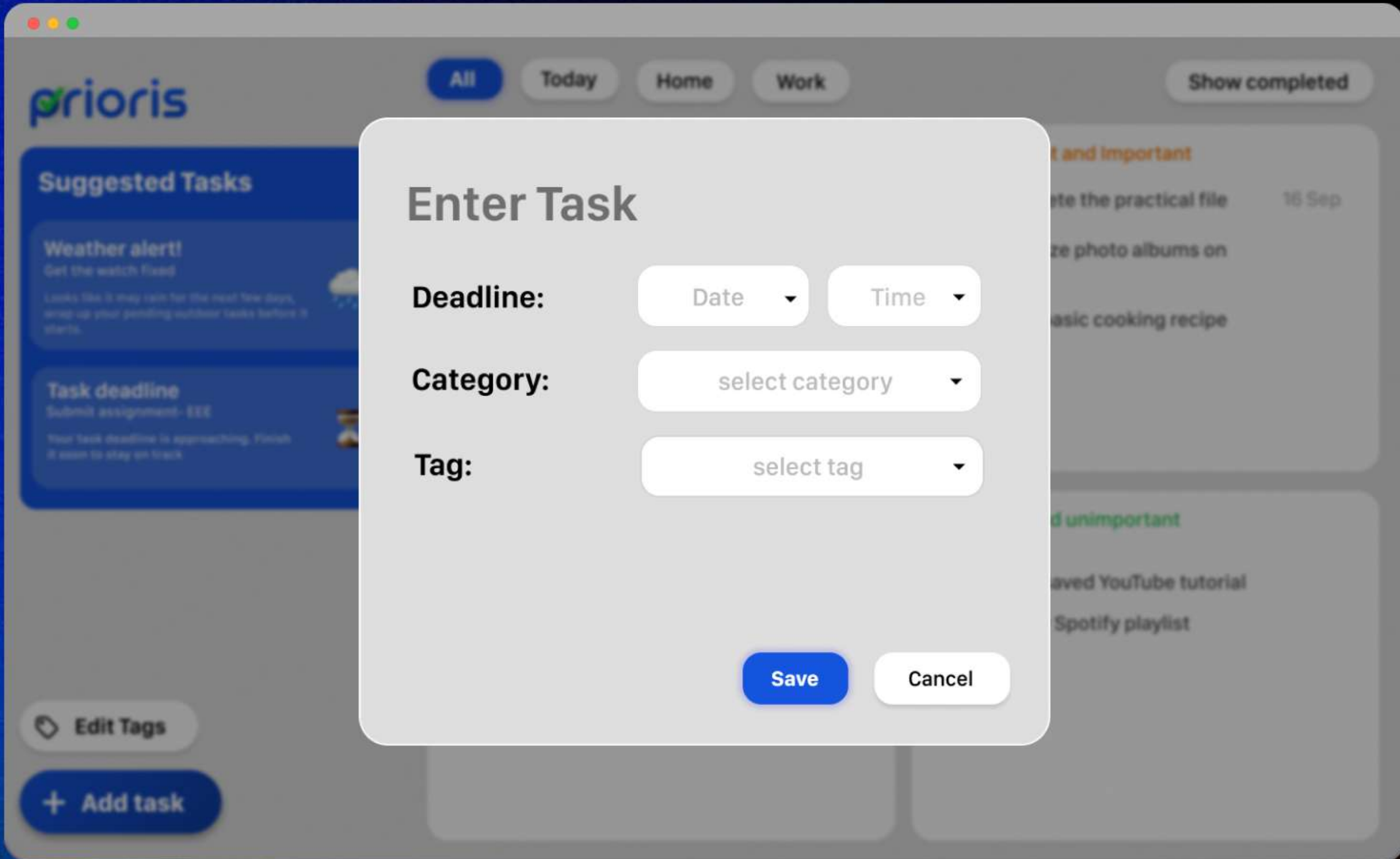
4 Urgent and unimportant

- ☐ Watch saved YouTube tutorial
- ☐ Try new Spotify playlist

User Interface



User Interface: creating new task



The image shows a screenshot of the 'prioris' app interface with a modal form titled 'Enter Task' overlaid. The background app shows a top navigation bar with tabs 'All', 'Today', 'Home', and 'Work', and a 'Show completed' button. On the left, there's a 'Suggested Tasks' section with items like 'Weather alert!' and 'Task deadline'. At the bottom left, there are buttons for 'Edit Tags' and '+ Add task'.

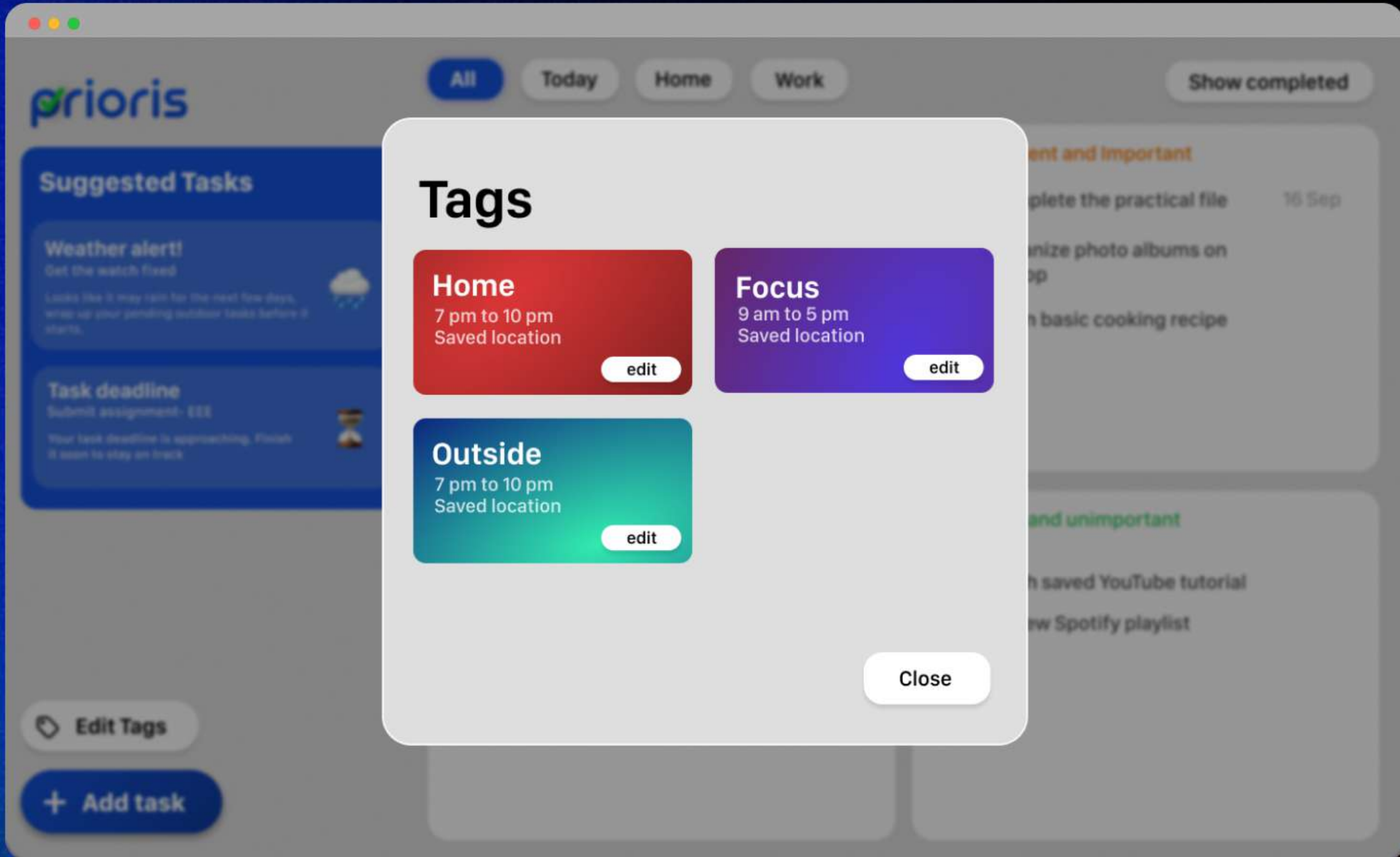
Enter Task

Deadline:

Category:

Tag:

User Interface: tags



Tech Stack

- Language → Python
- UI Framework → Tkinter
- Database → MySQL
- Context Modules →
 - Weather → OpenWeatherMap API
 - Location → Geopy
 - Focus/DND → system API calls

Advantages

- **Smarter prioritization with Eisenhower Matrix**
- **Context-aware reminders using tags (Home, Work, Commute, Focus)**
- **Suggested Tasks box shows only what's relevant now**
- **Reduced overload & distractions → better focus**
- **Timely notifications prevent missed deadlines**
- **Simple, minimal UI for quick task creation & management**

Future scope

- AI-based auto-prioritization of tasks from user habits
- Cross-platform support (Windows, macOS, Mobile)
- Calendar & email integration for seamless sync
- Voice commands & smart assistants for task input
- Cloud sync & multi-device access
- Productivity analytics & insights for better tracking



THANKS!

by:

Anubhav Sharma

Tanishk Bhatnagar

Ashish S Nair

RA2411026030010

RA2411026030018

RA2411026030022